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MM MEMO #VIBE-01

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To: Juha-Pekka Vierinen
From: Vibe implementation and validation record
Subject: **Vibe: SiO ablation port and validation**

1. Verdict: a useful scientific program, not yet a complete platform

Vibe successfully executed a substantially more demanding workload than Hello World or an FFT: coupled thermal ablation, adaptive integration, event detection, geodetic transformations, nested threshold searches, and parameter sweeps. The resulting program consists of **21 semantic functions** in one metablate.vibepack. The full run completed **521 study trajectories**, **22 cached boundary searches**, and **260 numerical assertions**. Search-internal trajectories and validation/refinement runs are additional to the 521. Eight study figures were produced in both PNG and SVG.

Conclusion. The task demonstrates that the extended Vibe bootstrap can express and execute a real numerical model accurately and quickly on a CPU. It does not establish full metablate compatibility, general numerical uncertainty certification, accelerator portability, large-project scaling, or improved LLM productivity. Those remain separate acceptance gates.

2. What was actually ported

The implemented scope is the Kero–Szasz thermal-ablation path used by the local ablatemm SiO studies, rather than every public API in the metablate package. The original source and saved study products were left unchanged.

Layer	Implementation and boundary
Physics	Vibe mass loss, heat balance, drag/scalar gravity, particle mass, local escape speed, radiation pressure and Poynting–Robertson quantities.
Numerics	Vibe Dormand–Prince 5(4), adaptive steps, dense event interpolation, mass/temperature objectives, bracket expansion and bisection.
Geometry / air	Vibe WGS84 conversions, fixed ray and log-density interpolation. NRLMSIS 2.1 supplies a fixed external input table; MSIS itself is not ported.
Host / evidence	Python allocates buffers, calls native entry points, stores HDF5, plots, and runs an independent original-RHS/SciPy reference. It supplies no ODE or optimization callbacks to the delivered Vibe studies.
Excluded	Alternative metablate models, plasma, sputtering, general dynamic transfer-coefficient machinery, and the complete Python package interface.

3. What Vibe had to learn

This was not a port onto an unchanged language. The compiler needed boolean comparisons, scoped `if/else` and `while`, typed scalar math intrinsics, checked numeric casts, and runtime bounds checks for indexed reads and writes. These were carried through the semantic API, type checker, query projections, precision traversal, import parser and C backend. The native build disables floating-point contraction and links platform libm.

The port is authored through a typed Python IR builder and atomic environment transactions, not through edited `.vibe` text. Generated C is disposable compiler output. JSON is transient tool transport, not a forest of persisted program files. This is a workable agent-facing route, but a bespoke authoring helper and manual ABI schema still expose missing standard-library/tooling work.

4. Physical model and numerical choices

Let m , v , s and T denote mass, speed, signed ray coordinate and temperature. With $A = 1.21(m/\rho_m)^{2/3}$, the implemented thermal terms are

$$\dot{m} = -4Ap_v(T)\sqrt{\frac{\mu}{2\pi k_B T}}, \quad p_v(T) = 0.1 \cdot 10^{C_A - C_B/T} \text{ Pa}, \quad (1)$$

$$mc_p \dot{T} = \frac{1}{2}\Lambda\rho_a v^3 A - 4\epsilon\sigma A(T^4 - T_0^4) + L\dot{m}, \quad (2)$$

$$\dot{v} = -\Gamma\rho_a v^2 A/m + GM/r^2, \quad \dot{s} = -v. \quad (3)$$

These reproduce the selected local implementation, including its scalar-gravity approximation; gravity is not projected along a dynamically curved orbit. The SiO inputs are $\rho_m = 2190 \text{ kg m}^{-3}$, $c_p = 848 \text{ J kg}^{-1} \text{ K}^{-1}$, $\mu = 44.0849 u$, $C_A = 14.841$, $C_B = 18755 \text{ K}$, $L = 359100/(44.0849 \times 10^{-3}) \text{ J kg}^{-1}$, $T_0 = 290 \text{ K}$, $\epsilon = 0.9$, and $\Lambda = \Gamma = 1$. Sensitivity sweeps explicitly vary one parameter at a time.

The atmosphere is the supplied 801-point NRLMSIS 2.1 profile at 250 m spacing from 0 to 200 km: Tromsø (69.5866°, 19.227°), 2018-06-28 12:45:33, $F_{10.7} = \bar{F}_{10.7} = 80$, $a_p = 4$. Stored float32 log densities are widened exactly to float64 before native interpolation. Zenith is measured at the 180 km starting altitude.

The state uses mass *fraction* to avoid applying a large absolute tolerance to a microscopic mass. Default relative tolerance is 10^{-7} ; absolute tolerances for $(m/m_0, v, s, T)$ are $(10^{-10}, 10^{-5}, 10^{-3}, 10^{-5})$ in the corresponding SI-normalized coordinates. Maximum step is 0.05 s. Integration stops at downward 400 K cooling, $m = \max(10^{-6}m_0, 10^{-18} \text{ kg})$, ground altitude, or 90 s. Events use quartic dense interpolation and bisection.

1350 K threshold speed (km/s)	$z = 0^\circ$	30°	60°	76.5552°
100 μm diameter	6.734	7.036	8.489	12.446
50 μm diameter	8.472	8.916	11.056	16.677

The last angle is the model's geometric straight-ray tangency, not a capture boundary. Threshold searches use 0.05 m/s root tolerance; this is a numerical stopping criterion, not physical precision. Peak temperatures are sampled at accepted states, not found by a continuous extremum solver.

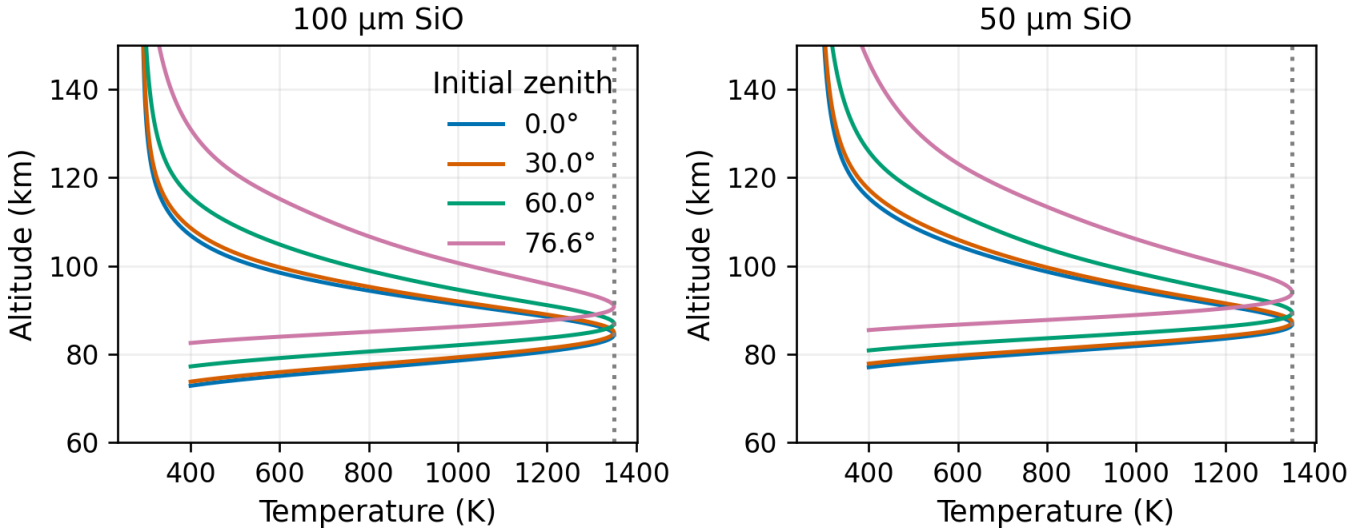


Figure 1: Vibe temperature profiles at the 1350 K thresholds. Curves show the fixed-ray model for four initial zenith angles.

Script: `scripts/report_metablate.py`; data: `scripts/run_metablate.py`.

5. Numerical validation: agreement, refinement, and old-output differences

The 260 numerical assertions comprise 195 kernel/coordinate/atmosphere/event checks, 63 trajectory/refinement checks (seven per case), and two radiation array comparisons. An assertion may compare many samples. Additional checks cover output-capacity failure and atomic rejection of a dimensionally invalid transaction. The compiler's 22 tests and shell smoke tests pass; a separate undefined-behavior-sanitized validation run also passes.

Nine trajectories were compared against *the original metablate RHS*, independently integrated with SciPy at matched component tolerances. They cover 50/100 μm particles, vertical/shallow entries, strong ablation, and the three largest relevant discrepancies found in saved studies. The comparison uses SciPy dense output at Vibe sample times within the common interval. Both share model constants and the input atmosphere, so this tests implementation agreement, not independent validation of the underlying physics.

Quantity	Largest observed difference	Acceptance criterion
Temperature	0.004755 K	0.1 K absolute
Mass fraction	1.3683×10^{-6}	2×10^{-5} absolute
Speed	0.006809 m/s	2×10^{-5} relative + 0.02 m/s
Signed distance	0.007791 m	2 m absolute
Terminal time	6.947×10^{-6} s	0.002 s absolute
Refined peak temperature	0.07950 K	0.15 K absolute
Refined retained fraction	4.875×10^{-7}	2×10^{-5} absolute

Refinement reduces relative/absolute tolerances by 100 and maximum step by four. These are **empirical error estimates**, not rigorous error bounds, and are not Vibe's automatic uncertainty-propagation feature. No claim of exhaustive parameter-space validation is made.

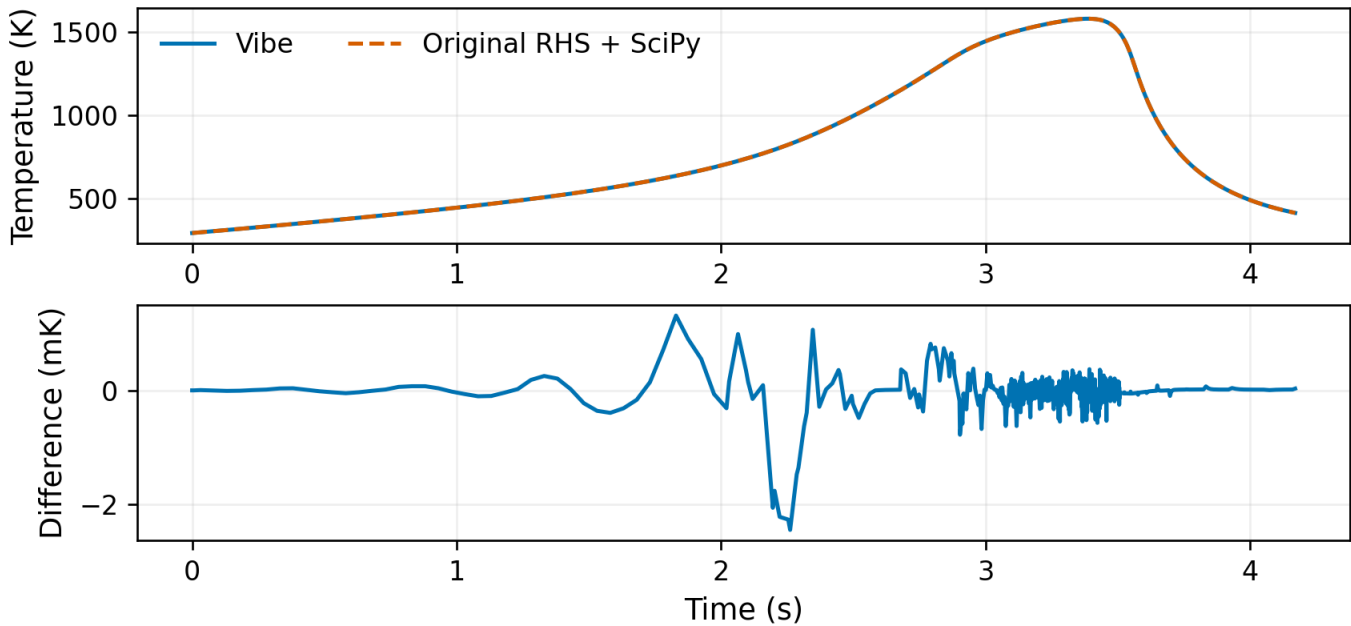


Figure 2: A challenging historical-discrepancy case: 20 μm SiO, 25.5 km/s, vertical entry. The tight original-RHS reference and Vibe agree; their temperature difference is shown in millikelvin below.

Script: scripts/report_metablate.py; data: scripts/run_metablate.py.

Saved loose-tolerance outputs are not exactly reproduced: the mass/velocity sweep differs by up to 25.8045 K and 0.0102712 in retained mass fraction; the emissivity sweep differs by up to 22.4817 K. Tight independent comparisons pass at those selected worst cases. The legacy solver used scalar absolute tolerance 10^{-6} even for microscopic SI masses, whereas this run resolves mass fraction. This supports a numerical-tolerance explanation, but is not an exhaustive attribution of every difference. The local Stefan–Boltzmann correction is preserved; the old distance-based ground event is corrected to actual WGS84 altitude. Historical deltas remain in HDF5 for inspection.

6. Reproduced scientific studies and interpretation limits

The run covers non-melt profiles and enhanced-speed comparisons (48 trajectories), survival boundaries and comparisons (18), the mass/velocity grid (357), escape-speed angle/size cases (18), and parameter sensitivity (80). The radiation-pressure/PR calculation adds 1201 native evaluations. Boundary searches include temperature, retained fraction, and parameter objectives; two extra cached searches arise from slightly different archived tangent angles.

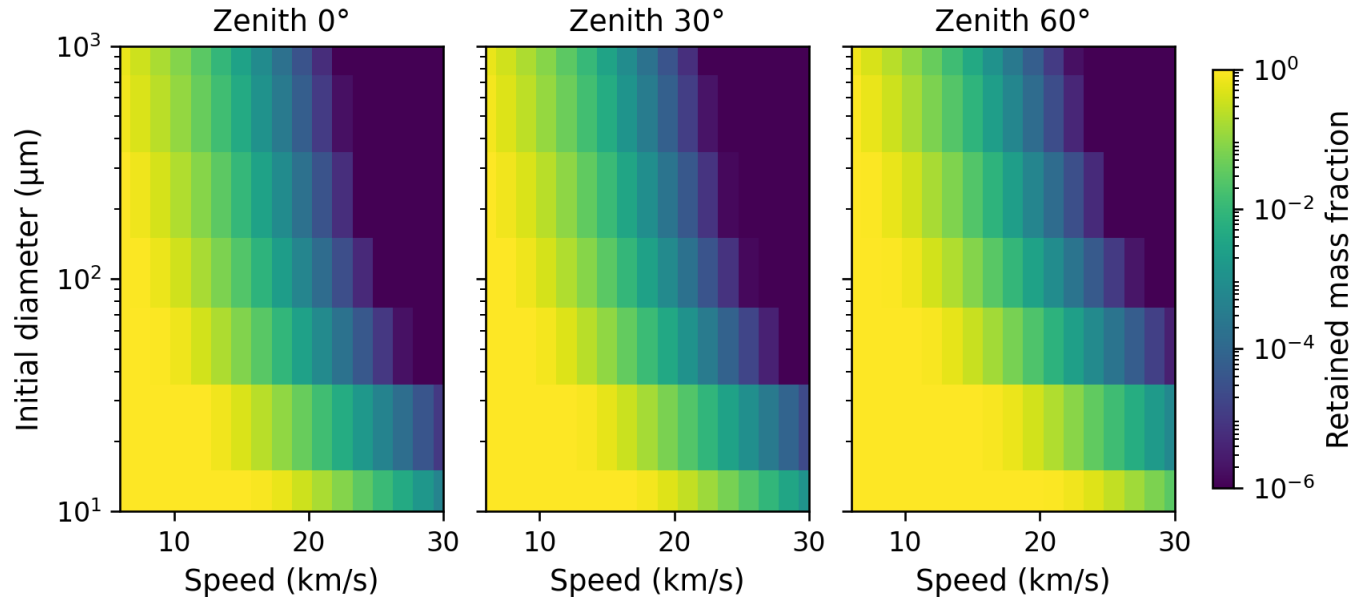


Figure 3: Retained mass fraction from the 357 Vibe trajectories, displayed on the sampled diameter/speed grid rather than as a fine-resolution physical boundary. The display floor is 10^{-6} ; unmodified numerical values are retained in HDF5.

Script: scripts/report_metablate.py; data: scripts/run_metablate.py.

100 μm survival speed (km/s)	1% retained	10% retained	50% retained
$z = 0^\circ$	16.125	12.871	9.737
$z = 30^\circ$	16.423	13.166	10.036
$z = 60^\circ$	17.904	14.596	11.460

These nine boundaries satisfy the study's 3000 K ceiling. They are thermal survival criteria in this model, not predictions of mineralogy or melt fraction. Smaller particles and shallower rays generally tolerate higher initial speed, but the finite grid is not evidence of behavior between every sampled point.

At the local escape speed, 11.04119 km/s, the vertical 100 μm sensitivity searches place the 1350 K crossing at $\Lambda \approx 0.23655$, $\Gamma \approx 4.05946$, or $\rho_m \approx 537.57 \text{ kg m}^{-3}$ when varied separately. These are diagnostic parameter variations, *not* adopted material values or a recommendation to increase drag artificially. The baseline remains $\Lambda = \Gamma = 1$.

What this study does not establish

There is no gravity-curved orbital propagation, capture history, coupled phase-transition model, material-parameter uncertainty propagation, or empirical atmospheric validation here. In particular, the tangency column in the threshold table cannot establish temporary capture or an aerocapture mechanism. Numerical agreement cannot make a simplified physical model more complete.

The generated study figures also include escape-angle profiles, sensitivity, and radiation pressure. They are available beside the consolidated HDF5 rather than all being reduced to unreadable thumbnails in this report. Report plots are generated at the actual 7.06-inch LaTeX width with 10–11 pt plot text.

7. CPU performance and engineering assessment

On this Apple M4 Pro (arm64, macOS 26.6.2, Apple Clang 21.0.0, Python 3.13.1), a warmed serial batch of the nine validation cases was timed three times, alternating implementation order. Compilation, atmosphere construction, plotting and reference resampling were excluded. Vibe timing includes host checks, trajectory allocation/copy and native integration; SciPy timing includes its original Python RHS and dense integration output. Component tolerances and maximum steps match, but the two adaptive solvers need not take identical steps.

Nine-case batch	Median wall time	Observed range
Compiled Vibe	8.608 ms	8.412–8.797 ms
Original RHS / SciPy	4.191 s	4.161–4.209 s

The median ratio is **487× for these implementations on this host**. This primarily demonstrates the value of keeping the repeated numerical work in compiled code instead of invoking the existing Python/object-heavy RHS. It is not a comparison with optimized NumPy, Numba, C++, or another native solver, nor a general language speed ranking. The full study/validation/plot phase took 9.04 s in its recorded run, excluding initial compilation and atmosphere setup; that mixed workflow time is not a kernel benchmark.

What worked. Typed semantic transactions produced a portable program; strict scalar types and checked SI interfaces caught invalid calls; explicit control flow expressed adaptive numerical work; native output reproduced independent references. A small runtime-checked numerical core was sufficient for a useful scientific application.

What was awkward. The implementation required compiler extensions and a custom Python authoring/ABI helper. State and configuration remain positional, mixed-quantity f64 arrays, not named dimension-typed records. The host checks mutable-buffer overlap, but there is no general ownership proof. Bounds failures terminate the native process rather than returning structured agent diagnostics. Peak extraction and tolerances are hand-designed.

What is unmeasured. This task does not measure LLM token consumption, first-pass validity, repair rate, concurrent-agent throughput, incremental rebuild cost or large-project scaling. The current pack still undergoes whole-pack validation/rewriting, and this host regenerates/compiles the whole program. GPU execution, AD, automatic uncertainty propagation and a production solver library are not implemented by this port.

8. Recommended next language work

Prioritize named records with per-field units and generated ABI bindings; reusable solver/event/result objects with structured failure evidence; semantic test and convergence records tied to exact revisions; and incremental function checking/code generation. Then benchmark agent editing/debugging against a Python baseline. Accelerator work should follow explicit array shapes, ownership and transfer semantics, not a promise inferred from CPU success.

9. Reproducibility and source record

From /Users/jvi019/src/vibec, after installing the original study’s dependencies in base conda:

```
cargo build
cargo test
bash scripts/test.sh
conda run -n base python scripts/run_metablate.py
conda run -n base python scripts/report_metablate.py
make -C reports/metablate
```

10. Artifact map and audit details

Executable semantic program

examples/metablate/metablate.vibepack: canonical program and semantic history. It can be checked or inspected with `vibec env`; no source-text reconstruction is needed to execute it.

Authoring and native host

scripts/author_metablate.py, scripts/semantic_builder.py, and scripts/metablate_native.py. The authoring script embeds RK45 coefficients from the installed SciPy table as constant semantic expressions. The host pins the semantic revision before C emission and ABI inspection.

Study runner and input/reference locations

scripts/run_metablate.py; original package: /Users/jvi019/src/ablate; study definitions and archived references: /Users/jvi019/src/ablatemm. The runner records source hashes and the package commit, including compiler dirty-worktree status rather than claiming an uncommitted compiler change is already reproducible from a Git commit alone.

Consolidated numerical product

/Users/jvi019/src/ablatemm/output_vibe/metablate_vibe.h5. Groups include atmosphere, nonmelt, survival, escape, sensitivity, radiation, validation and provenance. The record stores raw trajectories, tolerances, historical differences, independent reference samples, revision and native build information.

Report and timing evidence

reports/metablate/report.tex; figures and output/pdf/metablate/report_evidence.h5 are created by scripts/report_metablate.py. The latter stores all timing repetitions, host/compiler details, numerical maxima and a hash of the source HDF5.

Pinned program revision for the reported run

sha256:67d5b3c0be9bac23c631605faf2fc8dc6e3334642b9da9aedb5614e652011023

The exact source and generated-C hashes are retained in HDF5. The report's rounded tables describe this recorded run; regenerated measurements should be checked against the report before publishing a new revision.

Additional verification command

```
conda run -n base python scripts/run_metablate.py
--validate-only --sanitize --output target/metablate-sanitize
```

Join these two displayed lines for a shell invocation. This enables undefined behavior checks on the emitted native model. It supplements, rather than replaces, the numerical comparisons and compiler regression tests.

Source attribution and validation scope

Physics and coordinates are adapted from the specified metablate checkout, associated with Daniel Kastinen and Johan Kero, chiefly models/kero_szasz_2008.py, physics/thermal_ablation.py and coordinates.py. Study definitions are model_sio.py, model_sio_survival.py, memo001_nonmelt_profiles.py, model_sio_escape_angle.py, model_sio_parameter_sensitivity.py and radiation_pressure_pr.py. The derived example includes the upstream GPL license. This report is an implementation audit of those local sources, not a new literature review or an independent determination of SiO material properties.

Conclusion. Vibe did well as a compiled numerical substrate once the missing control-flow and math facilities were added. The strongest evidence is reproducible numerical agreement and a complete native study loop. The next challenge is making that result routine for agents: typed records, reusable numerical services, structured debugging evidence and measured incremental/concurrent development.
